

Applied Mathematics for Chemists

(화학수학)

Lecture 14

Operators



Partial Differential Equations

- General form of a linear second-order PDE

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \dots, G$ are functions of x and y .

- $G(x, y) = 0$: homogeneous
- $G(x, y) \neq 0$: inhomogeneous

Separation of Variables

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

Assume that $u(x, y) = X(x)Y(y)$.

$$\begin{aligned} X' &= dX/dx, X'' = d^2X/dx^2, \dots \\ Y' &= dY/dy, Y'' = d^2Y/dy^2, \dots \end{aligned}$$

If we obtain $f(x, X, X', \dots) = g(y, Y, Y', \dots)$, let

$$f(x, X, X', \dots) = g(y, Y, Y', \dots) = \lambda$$

where λ is the **separation constant**.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

Wave Equation (파동방정식)

From the equation of motion (Newton's law),

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\rho} \frac{\partial^2 y}{\partial x^2}$$

where T is the magnitude of the tension force and ρ is the mass of the string per unit length.

Letting $c = \sqrt{T/\rho}$,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Additional Conditions

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

- Boundary conditions

$$y(0, t) = y(L, t) = 0$$

- Initial conditions

$$y(x, 0) = 0, \quad \left. \frac{\partial y}{\partial t} \right|_{t=0} = f(x)$$

Solving Wave Equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Assume $y(x, t) = \psi(x)\theta(t)$;

$$\psi \frac{d^2 \theta}{dt^2} = c^2 \frac{d^2 \psi}{dx^2} \theta$$

$$\underbrace{\frac{1}{c^2 \theta} \frac{d^2 \theta}{dt^2}}_{\text{temporal part}} = \underbrace{\frac{1}{\psi} \frac{d^2 \psi}{dx^2}}_{\text{spatial part}} = \lambda$$

Final Solution

- Spatial solution

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

- Temporal solution

$$\theta(t) = b_2 \sin\left(\frac{n\pi ct}{L}\right)$$

- Total solution

$$y(x, t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi ct}{L}\right)$$

The Schrödinger Equation

- 1-dimensional time-dependent Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \mathcal{V}(x, t)\Psi$$

- Consider the case $\mathcal{V}(x, t) = \mathcal{V}(x)$!
- Separation of variables:

$$\underbrace{i\hbar \frac{\theta'(t)}{\theta(t)}}_{\text{depends only on } t} = -\underbrace{\frac{\hbar^2}{2m} \frac{\psi''(x)}{\psi(x)}}_{\text{depends only on } x} + \mathcal{V}(x)$$

depends only on t

depends only on x

Today's Class

- Operators
 - Operators
 - Operator algebra
 - Eigenvalue problem
 - Operators in quantum mechanics

13. Operators, Matrices, and Group Theory

(13장 연산자, 행렬, 군론)

Mathematical Operators (연산자)

perform **mathematical operations** on some function

e.g. "+3" operator, "×2" operator, "d/dx" operator

- Typically denoted by a letter with a **caret** (^)
- Operate on the function on its right

Some Examples

- **Multiplication operators** multiply a function by a constant or by another function
- **Derivative operators** differentiate a function one or more times
- **Symmetry operators** can move a point in space but can also operate on functions

Example 13.1

For the operator

$$\hat{A} = x + \frac{d}{dx},$$

find $\hat{A}f$ if $f = a \sin bx$, where a and b are constants.

Substitution gives

$$\hat{A}f = \left(x + \frac{d}{dx} \right) (a \sin bx) = ax \sin bx + ab \cos bx$$

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

Example 13.3

Find the operator equal to the operator product

$$\hat{A} = \frac{d}{dx} x.$$

Take an arbitrary function $f = f(x)$;

$$\hat{A}f = \frac{d}{dx}(xf) = f + x \frac{df}{dx} = \left(\hat{E} + x \frac{d}{dx} \right) f$$

Hence, the operator $\hat{A}$ is equal to

$$\hat{E} + x \frac{d}{dx}.$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A} ?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to **commute**
- **Commutator** (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Example 13.4

Find the commutator

$$\left[\frac{d}{dx}, x \right].$$

Take an arbitrary function $f = f(x)$;

$$\begin{aligned} \left[\frac{d}{dx}, x \right] f &= \left(\frac{d}{dx} x - x \frac{d}{dx} \right) f = \frac{d}{dx} (xf) - x \frac{df}{dx} \\ &= f + x \frac{df}{dx} - x \frac{df}{dx} = f = \hat{E} f \end{aligned}$$

Therefore,

$$\left[\frac{d}{dx}, x \right] = \hat{E}.$$

Rules for Commutativity

1. Operators acting on different independent variables commute

$$[x, d/dy] = 0$$

2. Multiplication and derivative operators usually do not commute
3. Two multiplication operators commute
4. Operator for multiplication by a constant commutes with any other operator

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A} \cdots \hat{A} \text{ (} n \text{ times)}$$

Example 13.5

If $\hat{A} = x + d/dx$, find $\hat{A}^2$.

$$\begin{aligned}\hat{A}^2 &= \hat{A}\hat{A} = \left(x + \frac{d}{dx}\right)\left(x + \frac{d}{dx}\right) \\ &= x^2 + \frac{d}{dx}x + x\frac{d}{dx} + \frac{d^2}{dx^2}\end{aligned}$$

Example 13.6

For the operator $\hat{A} = x + d/dx$, find $\hat{A}^2 f$ if $f(x) = \sin ax$.

From Example 13.5,

$$\hat{A}^2 = x^2 + \frac{d}{dx}x + x\frac{d}{dx} + \frac{d^2}{dx^2}$$

Substitution leads to

$$\hat{A}^2 f = \left(x^2 + \frac{d}{dx}x + x\frac{d}{dx} + \frac{d^2}{dx^2} \right) (\sin ax)$$

$$\hat{A}^2 f = x^2 \sin ax + \frac{d}{dx}(x \sin ax) + x \frac{d}{dx}(\sin ax) + \frac{d^2}{dx^2}(\sin ax)$$

$$= x^2 \sin ax + \sin ax + ax \cos ax + ax \cos ax - a^2 \sin ax$$

$$\hat{A}^2 f = x^2 \sin ax + \sin ax + 2ax \cos ax - a^2 \sin ax$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$
- Not all operators have inverses (*e.g.* $\hat{0}$)

Operator Algebra and Differential Equation

- Derivative operator $\hat{D}_x = d/dx$
- Homogeneous linear DE with constant coefficients

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

$$(a_n \hat{D}_x^n + a_{n-1} \hat{D}_x^{n-1} + \cdots + a_1 \hat{D}_x + a_0 \hat{E})y = 0$$

This can be solved by operator algebra

Example 13.7

Using operators, solve the equation

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0.$$

Using the operator notation,

$$(\hat{D}_x^2 - 3\hat{D}_x + 2\hat{E})y = 0$$

Noting that $[\hat{D}_x, \hat{E}] = 0$, factorization leads to

$$(\hat{D}_x - \hat{E})(\hat{D}_x - 2\hat{E})y = 0$$

Hence, the two roots are obtained from

$$(\hat{D}_x - \hat{E})y = 0$$

$$(\hat{D}_x - 2\hat{E})y = 0$$

Example 13.7

Using operators, solve the equation

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0.$$

$$(\hat{D}_x - \hat{E})y = 0, \quad (\hat{D}_x - 2\hat{E})y = 0$$

The two equations are identical with

$$\frac{dy}{dx} - y = 0, \quad \frac{dy}{dx} - 2y = 0$$

which respectively give

$$y_1 = e^x, \quad y_2 = e^{2x}$$

Hence, the general solution is

$$y = c_1 e^x + c_2 e^{2x}$$

Eigenfunctions and Eigenvalues

(고유함수와 고윳값)

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an **eigenfunction** of $\hat{A}$, and a is called an **eigenvalue**. This equation is called an **eigenvalue equation** (고윳값 방정식).

(Modified) Example 13.2

Find the eigenfunctions and eigenvalues for the operator d/dx .

We need to find a function that satisfies

$$\frac{df}{dx} = af$$

Solving this differential equation, one can get

$$f = Ce^{ax}$$

where C is an arbitrary constant.

The eigenfunction is $f = Ce^{ax}$, which corresponds to the eigenvalue a .

Operators in Quantum Mechanics

“For every **observable quantity**,
there is a corresponding **mathematical operator**”

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z)$$

Example 13.8

Find an eigenfunction of $\hat{p}_x$.

We need to find a function that satisfies

$$\hat{p}_x f = \frac{\hbar}{i} \frac{\partial f}{\partial x} = a f$$

The standard form is

$$\frac{\partial f}{\partial x} = \frac{ia}{\hbar} f$$

Solving this differential equation, one can get

$$f = C e^{iax/\hbar}$$

which is the eigenfunction of $\hat{p}_x$.

Hermitian Property (에르미트 성질)

- If $\hat{A}$ is **Hermitian**, it obeys the relation

$$\int \chi^* (\hat{A}\psi) dq = \int (\hat{A}\chi)^* \psi dq$$

where χ and ψ are arbitrary functions and q represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

Example 13.9

Show that the operator d/dx is not Hermitian (functions are bound).

$$\int \chi^*(\hat{A}\psi) dq = \int (\hat{A}\chi)^* \psi dq$$

Consider one dimension whose coordinate is x . If the operator is Hermitian, it obeys

$$\int_{-\infty}^{\infty} \chi^*(x) \left(\frac{d\psi}{dx} \right) dx = \int_{-\infty}^{\infty} \left(\frac{d\chi}{dx} \right)^* \psi(x) dx$$

Integration by parts of the left-handed side leads to

$$\int_{-\infty}^{\infty} \chi^*(x) \left(\frac{d\psi}{dx} \right) dx = \chi^*(x)\psi(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(\frac{d\chi}{dx} \right)^* \psi(x) dx$$

Example 13.9

Show that the operator d/dx is not Hermitian (functions are bound).

Need to show:

$$\int_{-\infty}^{\infty} \chi^*(x) \left(\frac{d\psi}{dx} \right) dx = \int_{-\infty}^{\infty} \left(\frac{d\chi}{dx} \right)^* \psi(x) dx$$

What we have:

$$\int_{-\infty}^{\infty} \chi^*(x) \left(\frac{d\psi}{dx} \right) dx = \overbrace{\chi^*(x)\psi(x)}^{\text{bound functions}} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(\frac{d\chi}{dx} \right)^* \psi(x) dx$$

Hence, the operator is not Hermitian.

Important Example

Show that the eigenvalues of a Hermitian operator are real.

For a Hermitian operator $\hat{A}$, the eigenvalue equation is

$$\hat{A}f = af$$

Multiplying both sides by f^* and integrating,

$$\int f^*(\hat{A}f) dq = \int f^*(af) dq = a \int |f|^2 dq$$

Since the operator is Hermitian, the left-handed side is

$$\int f^*(\hat{A}f) dq = \int (\hat{A}f)^* f dq = \int (af)^* f dq = a^* \int |f|^2 dq$$

Important Example

Show that the eigenvalues of a Hermitian operator are real.

$$a \int |f|^2 dq = a^* \int |f|^2 dq$$

Hence, if $\int |f|^2 dq \neq 0$,

$$a = a^*$$

which means that the eigenvalue a is real.